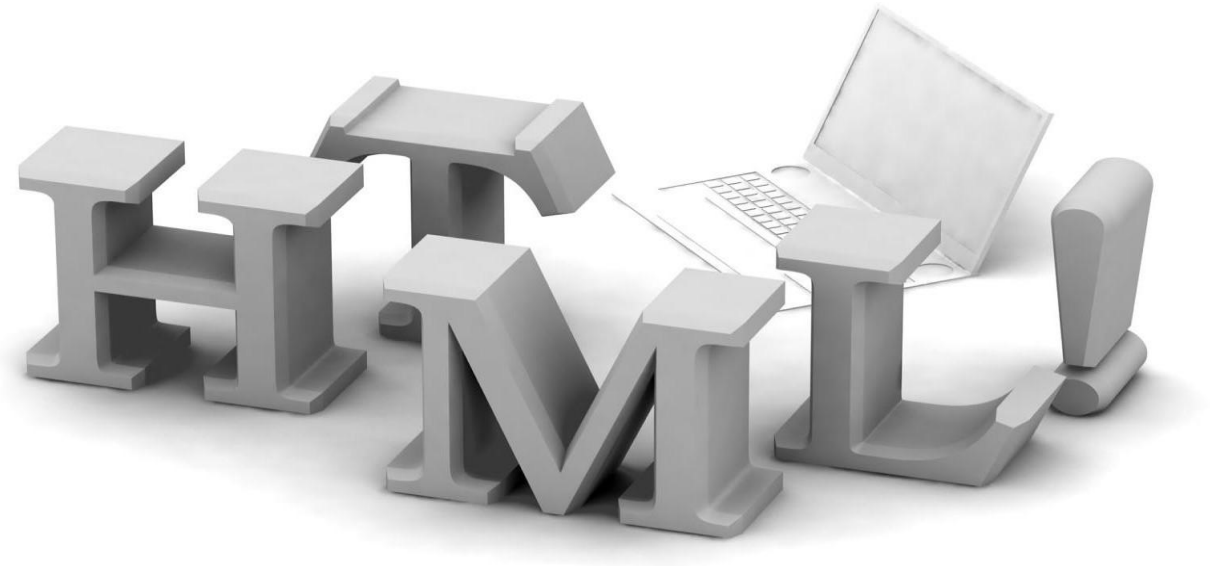
Faculty of Computer Science and Mathematics

Lab Exercise 2

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Overview:

This in-class exercise extends Lab Exercise 2. You will apply tables (including colspan and rowspan), forms (text inputs, radio buttons, checkboxes, drop-down lists and labels) and semantic HTML tags (header, nav, section, article and footer) to solve a realistic problem.

There are two problems. Your lecturer will assign one problem to you, or you may be asked to complete both.

Learning Outcomes

1. Create tables with headers and merged cells using colspan and rowspan.

2. Build forms with appropriate input types, correctly linked labels and grouped radio buttons.

3. Structure a web page using semantic HTML tags.

4. Identify and correct errors in tables, forms and semantic structure, and validate code using the W3C Validator.

Best of luck!

Problem A: Faculty Event Page with Schedule and Registration

Your faculty is organising a one-day event, for example a Programming Workshop, a Career Day or a Hackathon. The organising committee needs a simple web page that shows the event schedule and allows students to register. Build the page using HTML only (no CSS). You may choose any event name, date and activities.

Instructions

Create a file named lab2_ExerA.html that contains the following:

1. Semantic Structure

- A `<header>` containing an `<h1>` with the event name and a short tagline.
- A `<nav>` with internal links (e.g. `href="#schedule"`, `href="#register"`, `href="#about"`) that jump to each section of the page.
- At least three `<section>` elements, each with an `id`: About the Event, Event Schedule and Registration.
- Inside the About section, two `<article>` elements, each introducing one speaker or one activity (with an `<h3>` and a `<p>`).
- A `<footer>` with the organiser name and a copyright line using `©`.

2. Event Schedule Table

- A table with `border="1"` and header cells (`<th>`) for Time, Activity, Venue and Person in Charge.
- At least 6 rows of activities from morning to evening.
- Use `colspan` at least once (e.g. a "Lunch Break" row that spans across Activity, Venue and Person in Charge).
- Use `rowspan` at least once (e.g. two consecutive sessions held in the same venue).

3. Registration Form

Create a form with `action="register.html"` and `method="post"` that contains:

- Text inputs for Full Name, Matric Number and E-mail, each with a `<label>` whose `for` matches the input `id`.
- Radio buttons for Year of Study (Year 1 – Year 4). All radio buttons must share the same `name` so only one can be selected.
- Checkboxes for the sessions the student wants to join (at least 3 choices).
- A drop-down list (`<select>`) for Programme or Faculty with at least 4 options.
- A submit button.

4. Comments

- Use comments (`<!-- ... -->`) to explain each section of your code.

Submission

Save the file as lab2_ExerA.html, validate it at `https://validator.w3.org` (Validate by File Upload), and submit it (Submit lab2_ExerA.html and Screenshot of validation) through Github

Coding:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Faculty Technology Day</title>
</head>
<body>

    <!-- Header section: Event name and tagline -->
    <header>
        <h1>Faculty Technology Day 2026</h1>
        <p>Learn, Create and Innovate Through Technology!</p>
    </header>

    <!-- Navigation section: Links to page sections -->
    <nav>
        <a href="#about">About the Event</a> |
        <a href="#schedule">Event Schedule</a> |
        <a href="#registration">Registration</a>
    </nav>

    <!-- About section: Event information and activities -->
    <section id="about">
        <h2>About the Event</h2>

        <p>
            Faculty Technology Day is a one-day event organised to introduce
            students to programming, technology and career opportunities.
            Students can attend talks, participate in workshops and meet
            technology professionals.
        </p>

        <!-- Article 1: Speaker information -->
        <article>
            <h3>Guest Speaker: Introduction to Programming</h3>
            <p>
                Speaker: Dr. Muhammad Az'am. Learn the fundamentals of
                programming and discover how software is developed to
                solve real-world problems.
            </p>
        </article>

        <!-- Article 2: Activity information -->
        <article>
            <h3>Activity: Web Development Workshop</h3>
            <p>
                Join our hands-on workshop to learn how to create a simple
                website using HTML. Participants will build their first
                webpage with guidance from the instructors.
            </p>
```

```
52          </article>
53      </section>
54
55      <!-- Schedule section: Activities from morning to evening -->
56      <section id="schedule">
57          <h2>Event Schedule</h2>
58
59          <table border="1">
60              <thead>
61                  <tr>
62                      <th>Time</th>
63                      <th>Activity</th>
64                      <th>Venue</th>
65                      <th>Person in Charge</th>
66                  </tr>
67              </thead>
68
69              <tbody>
70                  <tr>
71                      <td>8:00 AM - 9:00 AM</td>
72                      <td>Registration and Breakfast</td>
73                      <td rowspan="2">Main Hall</td>
74                      <td>Event Committee</td>
75                  </tr>
76
77                  <tr>
78                      <td>9:00 AM - 10:00 AM</td>
79                      <td>Opening Ceremony</td>
80                      <td>Dean of Faculty</td>
81                  </tr>
82
83                  <tr>
84                      <td>10:00 AM - 11:00 AM</td>
85                      <td>Introduction to Programming</td>
86                      <td>Lecture Room 1</td>
87                      <td>Dr. Muhammad Az'am</td>
88                  </tr>
89
90                  <tr>
91                      <td>11:00 AM - 12:00 PM</td>
92                      <td>Web Development Workshop</td>
93                      <td>Computer Lab 1</td>
94                      <td>Ms. Fasya</td>
95                  </tr>
96
97                  <tr>
98                      <td>12:00 PM - 1:00 PM</td>
99                      <td colspan="3">Lunch Break</td>
100                 </tr>
101
102                 <tr>
```

```
                    <td>1:00 PM - 2:00 PM</td>
                    <td>Career Opportunities in IT</td>
                    <td>Seminar Room</td>
                    <td>Mr. Razali</td>
                </tr>

                <tr>
                    <td>2:00 PM - 3:00 PM</td>
                    <td>Student Project Showcase</td>
                    <td>Main Hall</td>
                    <td>Project Committee</td>
                </tr>

                <tr>
                    <td>3:00 PM - 4:00 PM</td>
                    <td>Closing Ceremony and Prize Giving</td>
                    <td>Main Hall</td>
                    <td>Event Committee</td>
                </tr>
            </tbody>
        </table>
    </section>

    <!-- Registration section: Student registration form -->
    <section id="registration">
        <h2>Event Registration</h2>

        <p>Please complete the form below to register for the event.</p>

        <form action="register.html" method="post">

            <!-- Text input for student's full name -->
            <label for="fullname">Full Name:</label>
            <input type="text" id="fullname" name="fullname" required>
            <br><br>

            <!-- Text input for matriculation number -->
            <label for="matric">Matric Number:</label>
            <input type="text" id="matric" name="matric" required>
            <br><br>

            <!-- Email input for student's email address -->
            <label for="email">E-mail:</label>
            <input type="email" id="email" name="email" required>
            <br><br>

            <!-- Radio buttons: All options share the same name -->
            <p>Year of Study:</p>

            <input type="radio" id="year1" name="year" value="Year 1" required>
            <label for="year1">Year 1</label>
```

```
            <br>

            <input type="radio" id="year2" name="year" value="Year 2">
            <label for="year2">Year 2</label>
            <br>

            <input type="radio" id="year3" name="year" value="Year 3">
            <label for="year3">Year 3</label>
            <br>

            <input type="radio" id="year4" name="year" value="Year 4">
            <label for="year4">Year 4</label>
            <br><br>

            <!-- Checkboxes: Students may select multiple sessions -->
            <p>Sessions You Want to Join:</p>

            <input type="checkbox" id="session1" name="sessions[]" value="Programming">
            <label for="session1">Introduction to Programming</label>
            <br>

            <input type="checkbox" id="session2" name="sessions[]" value="Web Development">
            <label for="session2">Web Development Workshop</label>
            <br>

            <input type="checkbox" id="session3" name="sessions[]" value="IT Career">
            <label for="session3">Career Opportunities in IT</label>
            <br>

            <input type="checkbox" id="session4" name="sessions[]" value="Project Showcase">
            <label for="session4">Student Project Showcase</label>
            <br><br>

            <!-- Dropdown list: Programme selection -->
            <label for="programme">Programme:</label>
            <select id="programme" name="programme" required>
                <option value="">-- Select Programme --</option>
                <option value="Software Engineering">Software Engineering</option>
                <option value="Computer Science">Computer Science</option>
                <option value="Information Technology">Information Technology</option>
                <option value="Data Science">Data Science</option>
                <option value="Cybersecurity">Cybersecurity</option>
            </select>
            <br><br>

            <!-- Submit button -->
            <button type="submit">Register Now</button>

            <!-- Reset button to clear the form -->
            <button type="reset">Clear Form</button>

        </form>
    </section>

    <!-- Footer section: Organiser and copyright information -->
    <footer>
        <p>Organised by the Faculty Technology Day Committee.</p>
        <p>&copy; 2026 Faculty Technology Day. All rights reserved.</p>
    </footer>

</body>
</html>
```

Problem B: Course Feedback Page (Debugging and Extend)

A student helper created a Course Feedback page for CSF3133. The page shows a feedback summary table and a feedback form, but it contains several errors. Some radio buttons can be selected at the same time, clicking some labels does nothing, and the table looks broken. Your tasks are to find and fix the errors, then extend the page.

Given Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8">
    <title>Course Feedback</title>
</head>
<body>
<header>
    <h1>CSF3133 Course Feedback</h1>
</head>
<nav>
    <a href="#results">Results</a> | <a href="#form">Feedback Form</a>
</nav>
<section id="results">
    <h2>Feedback Summary</h2>
    <table border="1">
        <tr>
            <th>Week</th>
            <th>Topic</th>
            <td>Rating</td>
        </tr>
        <tr>
            <td>1</td>
            <td>Introduction to HTML</td>
            <td>4.5</td>
        <tr>
            <td>2</td>
            <td colspan="3">Tables and Forms</td>
        </tr>
        <tr>
            <td>3</td>
            <td>Semantic HTML<td>4.0</td>
        </tr>
    </table>
</section>
<section id="form">
    <h2>Give Your Feedback</h2>
    <form action="submit_feedback.php" method="post">
        <label for="name">Name:</label>
        <input type="text" id="fullname" name="name"><br>
        <label>Overall rating:</label><br>
        <input type="radio" id="good" name="good" value="good">
        <label for="good">Good</label>
        <input type="radio" id="poor" name="poor" value="poor">
        <label for="poor">Poor</label><br>
        <input type="checkbox" id="lab" name="lab" value="yes">
        <label for="labs">I attended all labs</label><br>
        <input type="submit" value="Submit">
    </from>
</section>
<footer>
    <p>Copyright &copy 2026 CSF3133</p>
</footer>
</body>
</html>
```

Instructions

Part 1. Find and Fix the Errors

1. Copy the given code into a new file named lab2_ExerB.html.

2. Identify and correct all errors. At the top of the `<body>`, write an HTML comment listing each error and the line number, e.g. `<!-- Error 1 (Line 1): missing <!DOCTYPE html> -->`.

4. Validate your corrected file at `https://validator.w3.org`. Your final page must show "No errors or warnings to show".

Part 2. Extend into a Two-Page Website

5. Table: add rows for Week 4 and Week 5. Both weeks cover the same topic "Mini Project", so use `rowspan="2"` on the Topic cell. Add a final row "Average Rating" that uses `colspan="2".`

6. Form: add an E-mail text input with a matching `<label>`, a third radio option "Average", a group of at least 3 checkboxes "Topics I want more practice on", and a `<select>` for Section/Group (at least 3 options).

7. Semantic HTML: add a new `<section id="comments">` titled "Student Comments" that contains two <article> elements, each with an `<h3>` (student name) and a `<p>` (comment). Add a link to this section in the `<nav>`.

8. Use comments to explain each new section of your code.

Submission

Save the file as lab2_ExerB.html, validate it at `https://validator.w3.org` (Validate by File Upload), and submit it (Submit lab2_ExerB.html and Screenshot of validation) through Github

Coding:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>CSE3133 Course Feedback</title>
</head>
<body>

    <!--
    Part 1: Errors identified and corrected from the original code.
    Line numbers below refer to the original given code.

    Error 1 (Line 2): The html element was missing the correct
    language attribute. Corrected to lang="en".

    Error 2 (Line 5): The title element was not properly indented.
    Corrected for readability.

    Error 3 (Line 12): The navigation links used hbref instead
    of href. Corrected both links.

    Error 4 (Line 18): The Week table header used an incorrect
    th tag. Corrected the table heading structure.

    Error 5 (Line 20): The Rating heading used td instead of th.
    Corrected to th.

    Error 6 (Line 21): The table row structure was inconsistent.
    Corrected the opening and closing tags.

    Error 7 (Line 28): The colspan attribute was incorrectly
    applied to the Topic cell. Corrected the table structure.

    Error 8 (Line 30): The Week 3 row had incorrect table cells.
    Corrected the row structure.

    Error 9 (Line 38): The form action and method were incomplete
    for a real submission workflow. Kept a PHP action as an example.

    Error 10 (Line 39): The Name label had no matching for attribute.
    Added for="fullname" and a matching input id.

    Error 11 (Line 40): The input id and label association were
    incorrect. Corrected the id and name attributes.

    Error 12 (Lines 42-44): Radio buttons used different names,
    allowing multiple ratings. All now use name="rating".

    Error 13 (Line 45): The Poor label was incorrectly associated
    with the checkbox id. Corrected to for="poor".
```

```
    Error 14 (Lines 46-48): The checkbox label and input were
    incorrectly structured. Corrected their matching id attributes.

    Error 15 (Line 52): The copyright entity was written incorrectly.
    Corrected to &copy;.
    -->

    <!-- Header section -->
    <header>
        <h1>CSE3133 Course Feedback</h1>
        <p>Share your feedback to help us improve the course.</p>
    </header>

    <!-- Navigation: links to all main sections -->
    <nav>
        <a href="#results">Results</a> |
        <a href="#form">Feedback Form</a> |
        <a href="#comments">Student Comments</a>
    </nav>

    <!-- Part 1 and Part 2: Feedback summary table -->
    <section id="results">
        <h2>Feedback Summary</h2>

        <table border="1">
            <thead>
                <tr>
                    <th>Week</th>
                    <th>Topic</th>
                    <th>Rating</th>
                </tr>
            </thead>

            <tbody>
                <tr>
                    <td>1</td>
                    <td>Introduction to HTML</td>
                    <td>4.5/5</td>
                </tr>

                <tr>
                    <td>2</td>
                    <td>Tables and Forms</td>
                    <td>4.0/5</td>
                </tr>

                <tr>
                    <td>3</td>
                    <td>Semantic HTML</td>
                    <td>4.0/5</td>
```

```
                </tr>

                <!-- Week 4 and Week 5 cover the same Mini Project topic -->
                <tr>
                    <td>4</td>
                    <td rowspan="2">Mini Project</td>
                    <td>4.5/5</td>
                </tr>

                <tr>
                    <td>5</td>
                    <td>4.8/5</td>
                </tr>

                <!-- Final row: Average Rating spans two columns -->
                <tr>
                    <td colspan="2">Average Rating</td>
                    <td>4.4/5</td>
                </tr>
            </tbody>
        </table>
    </section>

    <!-- Part 2: Feedback form with email, rating, checkbox and dropdown -->
    <section id="form">
        <h2>Give Your Feedback</h2>

        <form action="feedback.php" method="post">

            <!-- Student name input -->
            <label for="fullname">Name:</label>
            <input type="text" id="fullname" name="fullname" required>
            <br><br>

            <!-- Student email input -->
            <label for="email">E-mail:</label>
            <input type="email" id="email" name="email" required>
            <br><br>

            <!-- Radio buttons share one name so only one can be selected -->
            <p>Overall Rating:</p>

            <input type="radio" id="good" name="rating" value="Good" required>
            <label for="good">Good</label>
            <br>

            <input type="radio" id="average" name="rating" value="Average">
            <label for="average">Average</label>
            <br>

            <input type="radio" id="poor" name="rating" value="Poor">
```

```
            <label for="poor">Poor</label>
            <br><br>

            <!-- Checkbox group: students can select multiple topics -->
            <p>Topics I want more practice on:</p>

            <input type="checkbox" id="html" name="topics[]" value="HTML">
            <label for="html">HTML</label>
            <br>

            <input type="checkbox" id="tables" name="topics[]" value="Tables">
            <label for="tables">Tables</label>
            <br>

            <input type="checkbox" id="forms" name="topics[]" value="Forms">
            <label for="forms">Forms</label>
            <br>

            <input type="checkbox" id="semantic" name="topics[]" value="Semantic HTML">
            <label for="semantic">Semantic HTML</label>
            <br><br>

            <!-- Dropdown: select a section or group -->
            <label for="group">Section/Group:</label>
            <select id="group" name="group" required>
                <option value="">-- Select Section/Group --</option>
                <option value="Group A">Group A</option>
                <option value="Group B">Group B</option>
                <option value="Group C">Group C</option>
                <option value="Group D">Group D</option>
            </select>
            <br><br>

            <!-- Submit and reset controls -->
            <button type="submit">Submit Feedback</button>
            <button type="reset">Clear Form</button>

        </form>
    </section>

    <!-- Part 2: Student Comments section with two article elements -->
    <section id="comments">
        <h2>Student Comments</h2>

        <!-- First student comment -->
        <article>
            <h3>Student: Abbas</h3>
            <p>
                The HTML lessons are useful and easy to understand.
                I would like more practical exercises.
            </p>
```

```
205             </article>
206
207             <!-- Second student comment -->
208             <article>
209                 <h3>Student: Faheem</h3>
210                 <p>
211                     The Mini Project helped me understand how to organise
212                     HTML elements and build a webpage.
213                 </p>
214             </article>
215         </section>
216
217         <!-- Footer section -->
218         <footer>
219             <p>&copy; 2026 CSE3133 Course Feedback</p>
220         </footer>
221
222     </body>
223     </html>
```

Rubric

Table A: Rubric for Additional Lab Exercise 2 – Problem A

Criteria	Description	Marks
Table Creation	Schedule table has <th> headers, at least 6 rows of meaningful data, and correct use of colspan and rowspan (no extra or missing cells).	3
Form Structure	Form contains text inputs with matching <label for> and id, a grouped set of radio buttons (same name), at least 3 checkboxes, a <select> with at least 4 options, and a submit button.	3
Semantic HTML	Proper use of <header>, <nav> with working internal links, <section> (min. 3), <article> (min. 2) and <footer>.	2
Code Organization and Readability	Code is well-organized, properly indented and commented. File is named lab2a.html and passes the W3C Validator.	2
Total		**10**

Table B: Rubric for Additional Lab Exercise 2 – Problem B

Criteria	Description	Marks
Error Identification	At least 7 errors correctly identified with line numbers and listed in an HTML comment.	3
Error Correction & Validation	All errors corrected. Radio buttons work as one group, all labels are linked, and the page passes the W3C Validator.	2
Table Extension	Week 4 and Week 5 rows added with correct rowspan; Average Rating row uses correct colspan. No extra or missing cells.	2
Form Extension	E-mail input with label, third radio option in the same group, at least 3 checkboxes and a <select> with at least 3 options.	2
Semantic HTML and Code Organization	New <section> with two <article> elements and a working <nav> link. Code is indented and commented.	1
Total		**10**